

All-Optical Wavelength Conversion

All-optical wavelength conversion – no electronics, too slow, too costly. All-optical is faster and more efficient

Applications: contention resolution, packet switching

Semiconductor optical amplifier (SOA) amplifies optical signals

Cross-gain modulation (XGM) and Cross-phase modulation (XPM) employ a nonlinear effect in the SOA called carrier depletion to convert wavelengths. Effectively, we modulate the information in our original signal onto another signal that is at our desired wavelength.

Four-wave mixing (FWM) is a nonlinear effect that occurs when two signals beat with each other, creating additional frequencies. What we can do is place a signal at just the right frequency/wavelength alongside our original signal, so that at the output, through FWM, we will have a copy of our original signal at our desired wavelength.

XGM, XPM, and FWM are fast, and can support 10 Gb/s